

Automated detection of HER2 gene copy number in breast cancer using deep learning techniques

Terisara Micaraseth^a, Shanop Shuangshoti^b, Sakdina Prommaouan^b, Somruetai Shuangshoti^c, Rizwan Ullah^d, Gridsada Phanomchoeng^{a,e,*}

^a Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok, 10330, Thailand

^b Department of Pathology, Faculty of Medicine, Chulalongkorn University, Bangkok, 10330, Thailand

^c The Institute of Pathology, Department of Medical Services, Ministry of Public Health, Bangkok, 10400, Thailand

^d School of Information Science and Technology, University of Science and Technology of China, Hefei, Anhui 230027, China

^e Center of Excellence in Applied Medical Virology, Department of Microbiology, Faculty of Medicine, Chulalongkorn University, Bangkok, 10330, Thailand

ARTICLE INFO

Keywords:

Breast cancer
HER2 DISH
Image processing
Deep learning
YOLOv11
Signal quantification
Brightfield microscopy

ABSTRACT

Accurate evaluation of HER2 gene amplification is critical for guiding breast cancer treatment decisions. This study proposes a deep learning-based diagnostic system for analyzing Dual In Situ Hybridization (DISH) images to support HER2 status assessment. The system integrates two models—YOLOv11-seg for cell detection and YOLOv11 object detection models for HER2 and CEP17 signal quantification—into a unified pipeline. High-resolution whole-slide images were preprocessed and annotated to train the models, which were then embedded into a standalone application designed for clinical environments. Upon uploading TIFF format images, the application performs automated cell detection, red/black signal analysis, and HER2/CEP17 ratio computation. Experimental results demonstrated an accuracy 95.24 % for the best identification and mean deviations of 6.08 % (CEP17) and 12.78 % (HER2) compared with manual counting. Statistical analyses confirm high consistency, particularly in red signal detection. Clinical feedback under scores the system's ease of use, accuracy, and potential to reduce diagnostic burden. The proposed approach demonstrates strong feasibility for routine adoption in pathology workflows.

1. Introduction

Breast cancer remains one of the most significant global health challenges, particularly among women. According to the World Health Organization (WHO), an estimated 19.3 million new cancer cases were diagnosed worldwide, reflecting the severe burden posed by cancer on healthcare systems and societies [1,2]. Among these, breast cancer constitutes a major cause of morbidity and mortality, where early and accurate detection is vital to improving clinical outcomes and guiding effective treatment strategies.

Breast cancer arises from complex genetic and molecular alterations in breast tissue. These changes may result from hormonal imbalances, inherited mutations, or acquired genetic disruptions that affect key oncogenes and tumor suppressor genes [3]. One aggressive subtype is human epidermal growth factor receptor 2 (HER2) positive breast cancer, accounting for approximately 20 % of all breast cancer cases [3].

HER2 overexpression correlates with increased tumor proliferation, higher recurrence risk, and greater metastatic potential—especially to the brain—often resulting in poor prognoses [4]. Accurate assessment of HER2 status is therefore critical in determining eligibility for HER2 targeted therapies, such as trastuzumab, which have significantly improved patient outcomes [5,6].

Clinicians rely on a range of imaging and molecular techniques to support diagnosis. These include sentinel lymph node biopsy, computed tomography (CT), magnetic resonance imaging (MRI), mammography, and positron emission tomography (PET), all of which help assess tumor presence, spread, and staging [7–10]. In parallel, molecular profiling through hormone receptor testing and HER2 evaluation guides personalized treatment planning [11,12]. For HER2 detection, in situ hybridization techniques—such as fluorescence in situ hybridization (FISH) and brightfield dual in situ hybridization (DISH)—allow direct visualization of gene amplification using color signal probes. In particular,

* Corresponding author. Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok, 10330, Thailand.

E-mail addresses: 6673004321@student.chula.ac.th (T. Micaraseth), Shanop@gmail.com (S. Shuangshoti), toytoysakdina@gmail.com (S. Prommaouan), somruetai.iop@gmail.com (S. Shuangshoti), rizwanustc@gmail.com (R. Ullah), gridsada.phanomchoeng@gmail.com (G. Phanomchoeng).

<https://doi.org/10.1016/j.ibmed.2025.100333>

Received 20 August 2025; Received in revised form 4 November 2025; Accepted 17 December 2025

Available online 18 December 2025

2666-5212/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

DISH enables the identification of black (HER2) and red (CEP17) signals under a conventional brightfield microscope, making it a practical alternative for many pathology labs.

In recent years, artificial intelligence (AI), and specifically deep learning (DL), has shown promising results in medical imaging analysis, offering high accuracy and automation in tasks such as tumor detection, classification, and treatment prediction [13–16]. Convolutional neural networks (CNNs) and object detection algorithms like You Only Look Once (YOLO) have been widely employed to localize pathological features in complex tissue images [17–20]. These methods have proven valuable in a range of cancer-related applications, including breast cancer detection, recurrence prediction, and HER2/FISH/DISH analysis [21,22].

While many DL-based methods have demonstrated excellent performance in image level or patch level classification, relatively few have focused on integrating these models into clinically useable systems for full slide HER2 DISH image analysis. Existing research often addresses only specific components, such as signal detection or segmentation, without connecting them to realworld usability or diagnostic workflows. In practice, manual analysis of HER2/CEP17 signals remains labor intensive and prone to human error, particularly in large whole-slide images with variable quality.

To address these gaps, an end to end prototype system was developed to assist in the analysis of HER2 gene amplification in brightfield DISH breast tissue images. Unlike previous works that emphasize model performance in isolation, this study focuses on creating a semi-automated assistive system that provides quantitative HER2/CEP17 signal analysis through a user friendly interface. To our knowledge, this is among the first implementations to combine model performance, nucleus-level quantification, and interface usability into a single deployable application, reducing manual workload while maintaining interpretability and consistency.

The key contributions of this work are as follows:

- An end to end prototype system was developed to assist in HER2 DISH image analysis, combining image preprocessing, deep learning-based cell detection, color signal quantification, and HER2/CEP17 ratio estimation within a unified workflow.
- The system incorporates a YOLOv11 seg-based model that demonstrates high capability in identifying clinically interpretable nuclei, thereby enabling reliable input selection for downstream signal analysis.
- The quantification module yields signal counts with accuracy levels considered acceptable in clinical settings, supporting its practical application as an assistive tool for pathologists analyzing HER2/CEP17 expression.

The sections of this research are organized as follows: Section II reviews related works in automated HER2 analysis and deep learning applications in pathology. Section III describes the materials and methods, including dataset preparation, image processing, and model development. Section IV presents the experimental results and discussion, highlighting key findings, practical implications, and limitations. Finally, Section V concludes the study and outlines potential directions for future work.

2. Related works

M. et al. integrates DL with explainable AI (XAI) techniques to enhance both the accuracy and transparency of brain tumor detection from MRI images. The approach employs a ResNet50 architecture in combination with Grad-CAM to generate heatmaps that highlight the most critical regions influencing the model's decisions. The model attained an accuracy of 98.52 % in distinguishing images containing tumors from normal images, with both precision and recall exceeding 98 %. These results demonstrate the potential of AI in supporting the

diagnosis of complex medical conditions [21]. Although the study focuses on brain tumor detection, its use of explainable AI (XAI) techniques highlights the potential to enhance interpretability in other medical image analysis tasks, including HER2 DISH evaluation. Zhu et al. (2024) focus on breast cancer detection using a model formed by integrating U-Net with Kolmogorov–Arnold Networks (KANs), referred to as U-KAN. This model demonstrated superior tumor boundary segmentation compared to the earlier U-Net++ architecture, achieving an accuracy of 93.3 % and a sensitivity of 97.4 %. The model was also tested on domain specific, high quality datasets such as BreastDM and BUSI, which are widely used in breast cancer research [18]. Ming et al. study presents CLAM, a deep learning framework that enables accurate cancer classification using only slide level labels, removing the need for detailed annotations. It demonstrates strong performance across multiple tasks, achieving AUC scores above 0.9 even with limited training data. CLAM generalizes well to independent test sets, biopsy slides, and cellphone microscopy images, showing its robustness and adaptability. The model provides interpretable attention heatmaps, helping identify key diagnostic regions without pixel level supervision. However, CLAM has not yet been validated for HER2 DISH slides, where precise signal detection and per nucleus pairing are essential. Overall, CLAM offers a data efficient, interpretable, and versatile solution for computational pathology in both clinical and research settings [23]. Hiroki et al. study successfully developed an automated BDISH assay for the simultaneous detection of HER2 gene amplification and chromosome 17 centromere (CEN 17) using a brightfield microscope. The assay demonstrated strong concordance with the FDA approved HER2 FISH test, offering a reliable alternative diagnostic tool. It enables pathologists to evaluate HER2 status using routine light microscopy, improving workflow efficiency and reducing the need for specialized fluorescence equipment. The clear visualization of black (HER2) and red (CEN 17) signals facilitates accurate assessment of HER2/CEP17 ratios. Overall, the BDISH method provides a practical, reproducible, and clinically applicable solution for HER2 testing in breast cancer diagnosis. While promising, this method does not include automated segmentation or real-time signal quantification, limiting its adaptability in fully digital pathology workflows [11].

Yang et al. study developed a multimodal deep learning model using ResNet50 to predict recurrence and metastasis risk in HER2 positive breast cancer patients. The model integrated H&E stained histopathological images with clinical data such as tumor stage and receptor status. On the internal training dataset (127 patients), the model achieved an AUC of 0.76, while on the independent TCGA test set (123 patients), it achieved an AUC of 0.72. The model also showed a sensitivity of 67 % and specificity of 83 %, outperforming predictions made using clinical data alone. These results suggest that combining image features with clinical information enhances predictive accuracy and could support more precise treatment decisions [24]. Nevertheless, the model targets outcome prediction rather than performing signal level HER2/CEP17 quantification. Wang et al. developed AWCDL, a deep learning model using Inception V4 with automatic weight calibration to detect HER2 amplification from IHC 2+ breast cancer slides. In internal validation, the model achieved 89 % accuracy, with equal sensitivity and specificity. External validation across two centers showed an average accuracy of 85 %, while the second step calibration improved accuracy to over 91 % in all centers. The model also successfully identified 50 % of HER2 positive cases among IHC 1+ slides, reducing false negatives. These results suggest that AWCDL is a reliable and cost effective alternative to traditional FISH testing [25]. However, AWCDL is limited to IHC 2+ slides and does not perform nucleus-specific signal analysis required in DISH evaluation. Pateel G P et al. studied HER2-ETNET, an ensemble deep learning model combining DenseNet-201, GoogLeNet, and ResNet-50 for multiclass classification of HER2 status (0+, 1+, 2+, 3+) using HE-stained breast cancer images. Transfer learning was applied, and model weights were assigned based on false positive and false negative rates, rather than empirical values. The proposed model

achieved 97.44 % accuracy, 97.32 % precision, 97.39 % recall, and AUC of 99.75 %, outperforming all baseline models and previous methods. Unlike conventional methods that depend on IHC, this approach can bypass expensive staining while maintaining high diagnostic accuracy. Yet, this method relies on HE-stained images and bypasses HER2/CEP17 signal detection altogether, making it unsuitable for DISH quantification. The results demonstrate that HE images contain sufficient information for reliable HER2 status classification using advanced deep learning techniques [14]. Rehman et al. study proposed a deep learning-based method for classifying HER2-SISH breast histopathology images into Normal, Amplified, and Non-Amplified regions. Four models were evaluated, with the Vision Transformer (ViT) achieving the highest accuracy of 100 % on internal patch level data. On external unseen WSIs, ViT maintained strong performance with 78 % accuracy, outperforming DenseNet201. The model demonstrated excellent agreement with expert annotations, supported by a Cohen's Kappa of 1.00. Overall, the approach offers a robust and automated solution to support pathologists in HER2 status assessment for breast cancer diagnosis [26]. Nonetheless, this approach classifies regions broadly and does not incorporate nuclear level pairing or signal count based scoring. Bappi et al. proposed a novel deep learning framework combining CNN, LSTM, and KNN with PCA for multi cancer classification. Two hybrid models, Vception (VGG + Inception) and Vmobilenet (VGG + MobileNet), were introduced. The best performance was achieved using an X-OR gate fusion, reaching 99.95 % accuracy for main classes and 99.13 % for subtypes. KNN with PCA showed superior results for Lymphoma, achieving 97.14 % accuracy. Overall, the model significantly improved diagnostic precision and reduced misclassification [27]. However, the model was designed for general cancer classification and lacks support for slide-level HER2/CEP17 signal analysis. Wang et al. study proposed a Soft Label Fully Convolutional Network (SL-FCN) to assist in breast cancer target therapy and thyroid cancer diagnosis. The model was tested on four datasets (DISH breast dataset 1 and 2, FISH breast dataset, and thyroid WSI) and was compared against 13 state of the art deep learning models including U-Net, FCN, YOLOv5, DeepLabv3+, and others. The SL-FCN achieved high accuracy across all datasets, notably 94.64 % on DISH dataset 2, 93.54 % on FISH dataset, and 99.99 % on the thyroid dataset, significantly outperforming baseline models ($p < 0.001$). It demonstrated strong segmentation performance even on complex images with blurry cell boundaries, thanks to its novel soft weight softmax loss and label smoothing. The results suggest SL-FCN can reduce diagnostic errors and enhance efficiency in HER2 status assessment and thyroid cancer detection [28]. Yet, the system does not provide a complete HER2 DISH workflow, including nuclear matching or automated HER2/CEP17 ratio computation.

Most prior models were designed for either classification or segmentation tasks, but none offer a complete, end-to-end system that is specifically clinically deployable for HER2 DISH analysis with signal-level precision and nuclear pairing. Previous studies lack nuclear-level HER2/CEP17 quantification, omit user-facing diagnostic tools, or are limited to IHC or fluorescence data. This work fills that gap by providing a full pipeline with per-nucleus signal detection, quantitative HER2 scoring, and an intuitive software interface for real-world use.

A comparison was conducted based on related works to evaluate the effectiveness of the proposed method and to highlight how well each approach can accurately classify the data as shown in Table 1.

In summary, although previous studies have addressed various stages of HER2 DISH image analysis—such as preprocessing, nucleus detection, signal localization, and HER2/CEP17 quantification—these components are often explored in isolation rather than as part of a cohesive diagnostic pipeline. Many methods rely on handcrafted preprocessing rules or threshold based signal detection, which are highly sensitive to staining variability and image quality. Nucleus detection models, though accurate at the patch level, are seldom integrated with downstream signal quantification or per-nucleus HER2 scoring. Furthermore, most approaches lack user interfaces, automation features,

Table 1

Comparative evaluation of deep learning approaches for breast cancer and HER2 assessment.

Ref	Model	Dataset	performance	Limitation
[24]	Multimodal deep learning (CNN + clinical data)	127 HER2-positive breast cancer patients (internal) and 123 TCGA patients (external)	0.76 (internal), 0.72 (TCGA)	Limited sample size, differences in race and experimental setup in TCGA data
[25]	AWCDL (Automatically Weighted Calibration Deep Learning)	IHC HER2 2+ images from 1073 patients across 3 cancer centers (376,099 tiles)	Internal: 89 %, External: 85 %	Still requires two-step validation; reduced accuracy in external centers; only applicable to HER2 2+ IHC cases
[14]	HER2-ETNET (Ensemble of DenseNet-201, GoogLeNet, ResNet50)	HE-stained breast histopathology images	Accuracy of test set 97.44 %	Evaluation is limited to one dataset; generalizability across varied clinical settings is untested.
[26]	Vision Transformer (ViT)	Private HER2-SISH WSIs (40 × magnification)	Patch level: 99.9 % Generalization: 78.8 % Avg (k-fold CV): 98 %	Limited generalization (only 78.8 %), complex image structure compared to H&E, private dataset
[27]	Vception (VGG + Inception), Vmobilenet (VGG + MobileNet), CNN-LSTM, KNN, SVM, X-OR Fusion	Secondary dataset from Kaggle: 130,000+ images, 8 primary cancer types, 26 subtypes (5000 images each)	Main class: 99.95 %, Subclass: 99.13 %; KNN (Lymphoma): 97.14 %	None explicitly stated; possibly reliant on secondary dataset and may require clinical validation
[28]	SL-FCN	DISH Breast Dataset 1, DISH Breast Dataset 2, FISH Breast Dataset, Papanicolaou-stained PTC WSIs	DISH1: 87.77 ± 14.97 %, DISH2: 94.64 ± 2.23 %, FISH: 93.54 ± 5.24 %, PTC: 99.99 ± 0.01 %	Performance varies with magnification; lower recall in DISH2; FISH affected by signal degradation; high variance in PTC precision/recall

or validation for clinical deployment, making them difficult to adopt in routine diagnostic workflows. To address these limitations, the present study proposes a clinically oriented, semiautomated prototype system that integrates deep learning based segmentation, signal detection, and HER2/CEP17 quantification into a single, useable platform. The system is designed to support pathologists in realworld settings by improving accuracy, reducing workload, and enabling scalable HER2 status assessment across diverse slide conditions.

3. Material and methods

This study was carried out with the formal approval and oversight of the Institutional Review Board (IRB), ensuring that all procedures involving human participants were conducted ethically, with careful attention to participant rights, safety, and confidentiality throughout the research process. The IRB approval was granted by the Institute of Pathology, Ministry of Public Health, Bangkok, Thailand (certification date: October 5, 2021). The titled “Multi Artificial Intelligence (AI)

Development in Digital Pathology'' (Project No. IOP-KM_R65-001).

3.1. Traditional cancer diagnostic methods

The diagnostic process for breast cancer using tissue samples typically begins with the collection of breast tissue, followed by staining techniques such as Dual In Situ Hybridization (DISH) to highlight specific genetic markers—most commonly HER2 and CEP17—using distinguishable colors. The HER2 DNA probe consists of a mixture of oligonucleotide probes that span approximately 300,000 base pairs within the genomic region containing the HER2 gene on Chromosome 17. The CEP17 probe targets sequences in the centromeric region of Chromosome 17 and serves as a reference for detecting aneusomy.

In stained cells, HER2 gene amplification is visualized as black points, while CEP17 signals appear red. The number of HER2 and CEP17 signals are counted within tumor nuclei to assess gene amplification status.

Next, stained slides are imaged using high resolution microscopic photography, producing digital images for evaluation. Then, pathologists review these images to detect cellular abnormalities and assess HER2/CEP17 signal patterns. Finally, the HER2/CEP17 signal ratio is calculated and reported, which informs diagnostic conclusions and guides subsequent clinical decisions.

3.2. The proposed diagnostic methods

To address the limitations of manual HER2/CEP17 signal evaluation described in Section 3.1—such as observer variability, labor intensity, and inconsistency across large tissue slides—this study proposes a semiautomated diagnostic pipeline that leverages deep learning for HER2 DISH image analysis. The process begins with conventional DISH

staining and high resolution brightfield microscopy, followed by automated image processing steps. These include image cropping to isolate regions containing nuclei, algorithmic selection of diagnostically suitable cells, and signal detection to quantify HER2 (black points) and CEP17 (red points) within each selected nucleus. The computed HER2/CEP17 ratio is used as a molecular diagnostic marker to assess HER2 gene amplification. Unlike previous approaches that focus solely on classification or segmentation, the proposed method integrates all critical steps into a single pipeline designed for clinical feasibility. By automating key steps in the image analysis process, the proposed pipeline helps reduce human error, minimize inter-observer variability, and alleviate the diagnostic workload for pathologists. This results in more consistent and objective assessments of HER2 gene amplification. Moreover, by enhancing both efficiency and accuracy at the molecular level, the system is well suited for integration into routine clinical workflows and computer aided diagnostic (CAD) systems, particularly in high volume pathology settings.

The complete diagnostic workflow—contrasting the traditional manual method and the proposed AI-assisted system—is illustrated in Fig. 1. In the traditional approach (Fig. 1a), pathologists are responsible for the entire process: preparing stained slides, visually examining them under a microscope, manually counting HER2 (black) and CEP17 (red) signals, and generating diagnostic reports. This method is time-consuming and prone to variability. In contrast, the proposed approach (Fig. 1b) shifts the analytical burden to an automated system. After tissue staining and whole-slide image (WSI) scanning, the deep learning models handle nucleus detection, signal quantification, and HER2/CEP17 ratio computation. The pathologist's role is streamlined to preparing the tissue and reviewing the AI-generated report, thereby reducing workload and enhancing diagnostic consistency.

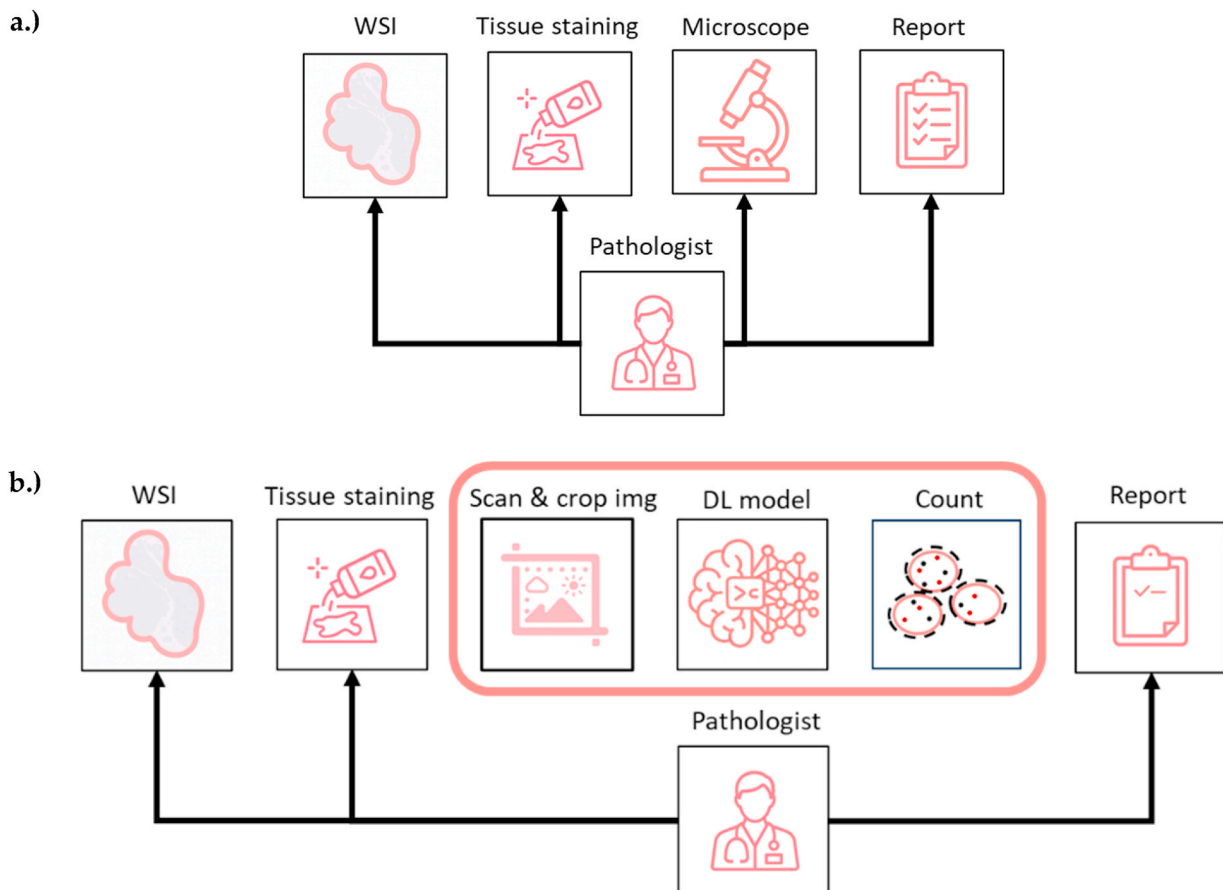


Fig. 1. Overview of the diagnostic workflow. a) Traditional Cancer Diagnostic Methods and b) Proposed procedure diagnostic Methods.

3.3. The staining process

The detection of HER2 gene amplification in this study follows the standard Dual In Situ Hybridization (DISH) protocol. This staining method, typically performed by a pathologist or laboratory technologist, uses chromogenic reactions to visualize HER2 and CEP17 gene regions under a brightfield microscope. Specifically, black signals indicate HER2 amplification, while red signals highlight the centromere of chromosome 17 (CEP17). In Fig. 2, illustrates the step by step chemical staining process involved in this protocol.

HER2 gene amplification is traditionally assessed by manually counting black (HER2) and red (CEP17) signals under a light microscope. Diagnostic classification is based on two key parameters: the HER2/CEP17 ratio and the average HER2 signal count per cell. According to established clinical guidelines, a case is considered positive if the HER2/CEP17 ratio is ≥ 2.0 and the average HER2 copy number is ≥ 4.0 per cell and counting at least 20 representative nuclei per case in HER2 ISH [29]. Cases that fall below both thresholds are classified as negative. These criteria, summarized in Table II, are essential for identifying HER2 overexpression, which is closely linked to aggressive breast cancer subtypes. Accurate determination of HER2 status is therefore critical for guiding targeted therapy decisions, particularly in selecting patients eligible for treatments such as trastuzumab.

Manual interpretation of HER2 and CEP17 signals is time-consuming, labor intensive, and prone to both intra- and inter observer variability. These limitations can lead to inconsistent assessments and potentially suboptimal treatment decisions. To address this, improving the reliability and efficiency of HER2 evaluation has become a critical objective in clinical pathology. Automated image analysis systems—particularly those powered by artificial intelligence and deep learning—offer a promising solution by increasing accuracy, reducing subjectivity, and accelerating diagnostic workflows. Such systems are designed not to replace pathologists, but to assist them in delivering faster, more consistent, and reproducible assessments.

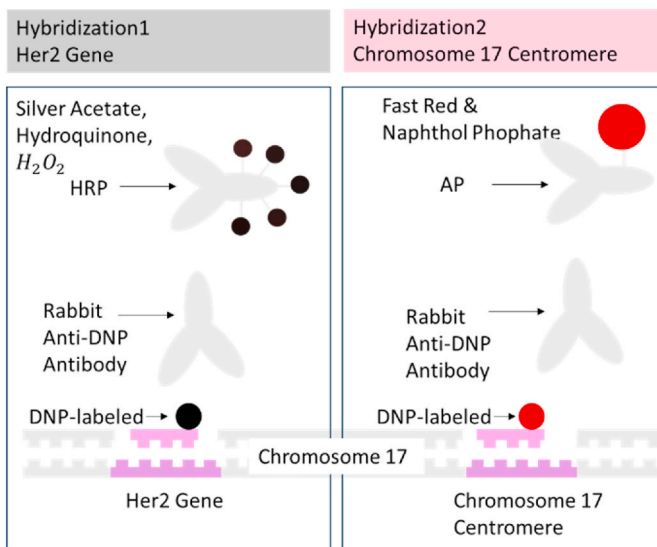


Fig. 2. Sequential in Situ Hybridization Method for brightfield Dual in situ Hybridization [11].

Table 2

Interpretation of dual in situ hybridization (DISH).

Classification	HER2/CEP17 Ratio	Avg HER2 Copy Number	Interpretation
Positive	≥ 2.0	≥ 4.0 signals per cell	Indicates HER2 gene amplification
Negative	< 2.0	< 4.0 signals per cell	Indicates no significant HER2 gene amplification

As summarized, the image processing pipeline—from color separation to point counting—plays a key role in enhancing the efficiency of HER2 and CEP17 signal analysis. Manual interpretation of these signals requires high precision and is prone to error. By automating these critical steps, the proposed system reduces diagnostic workload, shortens processing time, and improves overall accuracy and consistency. This ultimately enhances the reliability of HER2 status evaluation and supports its clinical applicability.

3.4. Dataset

In Fig. 3a, the whole-slide images (WSIs) of breast tissue samples were acquired using a brightfield microscope at Chulalongkorn University and initially saved in VSI format. These high resolution images were subsequently processed using Olyvia software (version 4.0), where each file was cropped to reduce its size to below 4 GB, resulting in partial whole-slide images (Figs. 3b–1). To facilitate efficient data handling and enable precise image analysis, the partial WSIs were further divided into smaller tiles of 1500×1500 pixels, referred to as intermediate images (Figs. 3b–2). These images were saved in TIFF format to preserve signal clarity through lossless compression. During the cropping process, a 10 % overlap was applied between adjacent image tiles to minimize the likelihood of excluding regions of interest near image borders. This tiling approach ensures that cellular structures such as nuclei and gene signals remain fully visible within each frame, which is critical for downstream analysis tasks. The intermediate image size was selected to balance annotation convenience, visual interpretability, and compatibility with the deep learning pipeline described in Section 3.5. A summary of the number of images generated at each processing stage—whole-slide, partial, and intermediate is provided in Table III.

3.5. Deep learning-based analysis of HER2 gene amplification in Dual In Situ Hybridization (DISH) images

To address the limitations of manual signal interpretation in HER2 DISH analysis, this section outlines the development of a deep learning-based approach that automates the detection and quantification of HER2 gene amplification. The overall development process, summarized in Fig. 3, consists of two main components: (1) cell segmentation to localize individual nuclei, and (2) HER2/CEP17 signal detection to determine gene amplification status. Each stage of the model development is described in detail in Sections 3.5.1. and E2.

3.5.1. Segmentation

The segmentation module was developed to accurately localize individual nuclei in DISH stained tissue images. The training process consisted of two main stages: annotation and model training.

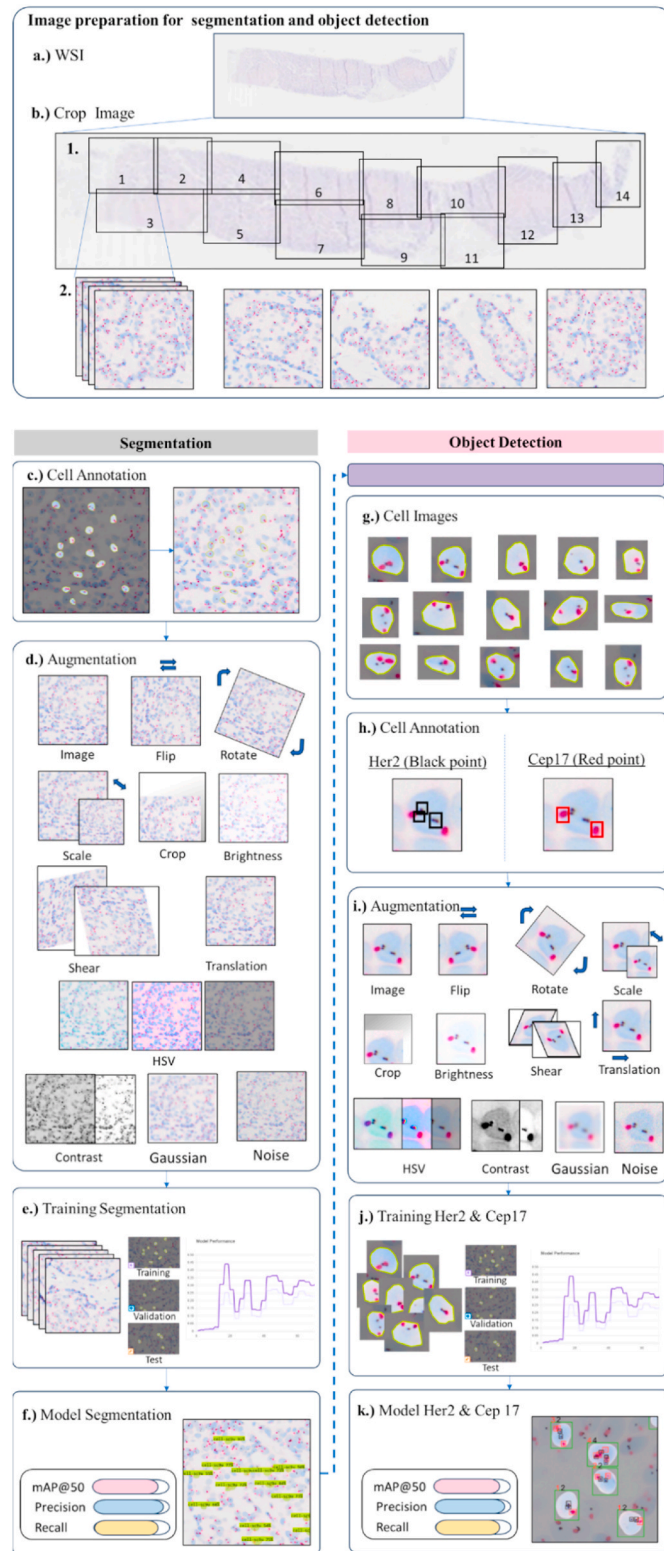


Fig. 3. Overview of model development pipeline for HER2 and CEP17 detection.

Table 3

Summary of dataset: number of images at each processing stage.

WSI	Partial whole slide images	Intermediate images
5	72	14,400

1) Annotation cells

The training of DL models for HER2 DISH detection was facilitated by a dedicated annotation process, in which ground truth cell annotations were performed by two certified pathologists. High resolution cropped tissue images were uploaded to the Roboflow platform, where bounding boxes were manually drawn around individual nuclei that met strict inclusion criteria—non-overlapping, fully visible, and containing distinguishable red (CEP17) and black (HER2) signals. Additional annotations were created for individual signal points within the nuclei, separating red and black signals into distinct datasets. The resulting labeled datasets were exported in YOLO-compatible format and were used in separate training processes for nucleus detection and color signal quantification. This precise labeling ensured high quality training data for all models in the pipeline.

2.) Cell detection using specialized YOLOv11-seg Models

Cell detection using specialized YOLOv11-seg models High-quality annotated data were used to train the YOLOv11-seg model for detecting diagnostically relevant nuclei in HER2 DISH tissue images. The training was performed on the Google Colab platform, leveraging image datasets previously labeled on Roboflow. These labeled images comprising well-isolated nuclei with clear HER2 and CEP17 signals were imported into the training environment.

Three YOLOv11-seg configurations with Fast, Medium, and Large were developed to evaluate trade-offs between model size, inference speed, and detection accuracy. The Fast model focuses on rapid processing and lightweight deployment, the medium model offers balanced performance, and the large model achieves the highest detection precision. Their comparative performance was analyzed to identify the most suitable configuration for clinical-grade HER2 DISH cell detection and ratio computation.

The YOLOv11-seg model was selected due to its proven performance to detect the cell and its ability to identify small, densely packed biological structures. Hyperparameters were optimized to maximize sensitivity while minimizing false positives, thereby ensuring reliable detection of nuclei across varied tissue patterns. The data split in to 70:20:10. This trained model represents the first detection stage in the overall analysis pipeline.

To improve generalization and prevent overfitting, image augmentation techniques were applied during training. The augmentation configuration, including geometric and photometric transformations, is summarized in Table IV.

3.5.2. Object detection

1) Annotation of Her2 & Cep17 Signals

Table 4

Image augmentation for Segmentation models.

Augmentation	Parameter	Value
Flip	Horizontal	50 %
	Vertical	50 %
Rotation	Max Rotation Angle	15°
Scale	Scale Range	1.0
Shear	Shear Angle Range	10°
Translate	X and Y Translation	10 %
Crop	Random Crop (relative size)	0.8
Brightness	Brightness Delta	0.2
Contrast	Contrast Range	1.2
Saturation	Saturation Range	1.2
Hue Shift	Hue Range	15°
Gaussian Blur	Kernel Size	5 px
Add Noise	Noise Amount	0.05
Cutout/Occlusion	Max Area Size	20 %

Following nucleus segmentation, the identified nuclear regions were cropped to isolate individual cells for molecular analysis. Each cropped image was evaluated against strict quality criteria, including visibility, completeness, and staining clarity. Nuclei that failed to meet these standards—such as those that were partially visible, overlapping, or poorly stained were excluded from further processing. The remaining high quality images were then annotated with distinct red and black points, representing CEP17 and HER2 signals by two certified pathologists. These annotations formed the basis for training signal detection models.

2) Color Detection Using Specialized YOLOv11 Models and RF-DETR model

Accurate quantification of HER2 (black) and CEP17 (red) signals was achieved using datasets derived from the best-performing segmentation models. The top 100 images were selected from the segmentation results, containing clearly distinguishable red and black signals within cell regions to enhance model sensitivity and precision. Independent YOLOv11 and RF-DETR architectures were trained for each color target, with Fast and Medium sub-model variants tailored for specific objectives. The data split in to 70:20:10. The Fast variants prioritized spli detection and deployment efficiency, whereas the Medium variants emphasized higher accuracy and deeper feature extraction. Developing multiple models allowed targeted optimization for different diagnostic requirements—balancing inference speed, precision, and robustness across diverse staining and imaging conditions.

This dual-model strategy enabled accurate computation of HER2/CEP17 signal ratios and facilitated semi-automated classification aligned with clinical diagnostic guidelines. Robustness and generalizability were further improved through extensive image augmentation, including geometric transformations (e.g., flipping, rotation, translation), photometric adjustments (e.g., brightness, contrast, hue), and noise injection methods (e.g., Gaussian blur, cutout). The full augmentation parameters are summarized in Table V.

3.6. Application

To facilitate and reduce the diagnostic workload for medical professionals, this study integrated all image processing

functions—including the YOLOv11 models—into a single standalone application. The system was developed to execute the full analytical pipeline, from image upload and preprocessing to automated cell detection, signal quantification, and HER2/CEP17 ratio computation, in a cohesive manner. The user interface (UX/UI) was designed to be simple and intuitive, minimizing the learning curve and eliminating the need for programming knowledge or manual parameter tuning, thereby making it suitable for routine clinical use.

The diagnostic process begins with the user entering a unique patient identifier (e.g., c-001) into a designated input field. This identifier is used to associate diagnostic results with specific patients and allows for future retrieval through electronic medical records. After entering the ID, users can upload a TIFF format image of a stained cell slide. The upload mechanism supports drag and drop functionality as well as manual file selection via a “Browse files” button. The system accepts image files up to 5 GB in size, accommodating high resolution microscope captures necessary for accurate DISH analysis.

Upon file upload, the application automatically initiates image processing. This includes calling the YOLOv11-seg model for cell detection, followed by two specialized YOLOv11 models that quantify the red (CEP17) and black (HER2) signal points within each detected nucleus. Once the analysis is complete, a notification is displayed to the user indicating successful processing (e.g., “Image processing complete for Patient ID: c-001”), and the diagnostic results are rendered in both numerical and visual formats.

The upper portion of the results panel displays key quantitative metrics, including the sum of HER2 (black) and CEP17 (red), the HER2/CEP17 signal ratio, and the average HER2 signal count per cell. These indicators are essential for assessing HER2 gene amplification status based on clinical guidelines.

Below the summary, the system displays the original slide image, overlaid with green bounding boxes generated by the YOLOv11-seg model. These boxes highlight the selected nuclei that meet the pre-defined criteria—clearly visible, nonoverlapping, and containing interpretable signal pairs. Each bounding box is annotated with the number of detected CEP17 (red) signals within the respective cell. Users can zoom and pan through the image using standard mouse controls, facilitating visual inspection and manual confirmation if needed.

By integrating all required functions into a unified and easy to use platform, the system effectively reduces diagnostic complexity, increases throughput, and enhances consistency in HER2 status evaluation. This approach supports the incorporation of artificial intelligence in clinical workflows and demonstrates the feasibility of deploying deep learning-based decision support tools in real world diagnostic environments.

The user interface, illustrated in Fig. 4, The system workflow begins with the upload of a high resolution cell image, which is then processed through segmentation to isolate individual cells. HER2 and CEP17 signals are subsequently detected and quantified. The interface features a Patient ID input field for linking diagnostic results to electronic medical records, a search function for retrieving prior analyses by Patient ID, and a quantitative summary panel displaying signal counts and ratios. This streamlined layout enhances user efficiency, minimizes manual interactions, and is well suited for high demand clinical environments.

Table 5
Image augmentation for object detection models.

Augmentation	Parameter	Value
Flip	Horizontal	50 %
	Vertical	50 %
Rotation	Max Rotation Angle	15°
Scale	Scale Range	1.0
Shear	Shear Angle Range	10°
Translate	X and Y Translation	10 %
Crop	Random Crop (relative size)	0.8
Brightness	Brightness Delta	0.2
Contrast	Contrast Range	1.2
Saturation	Saturation Range	1.2
Hue Shift	Hue Range	15°
Gaussian Blur	Kernel Size	5 px
Add Noise	Noise Amount	0.05
Cutout/Occlusion	Max Area Size	20 %

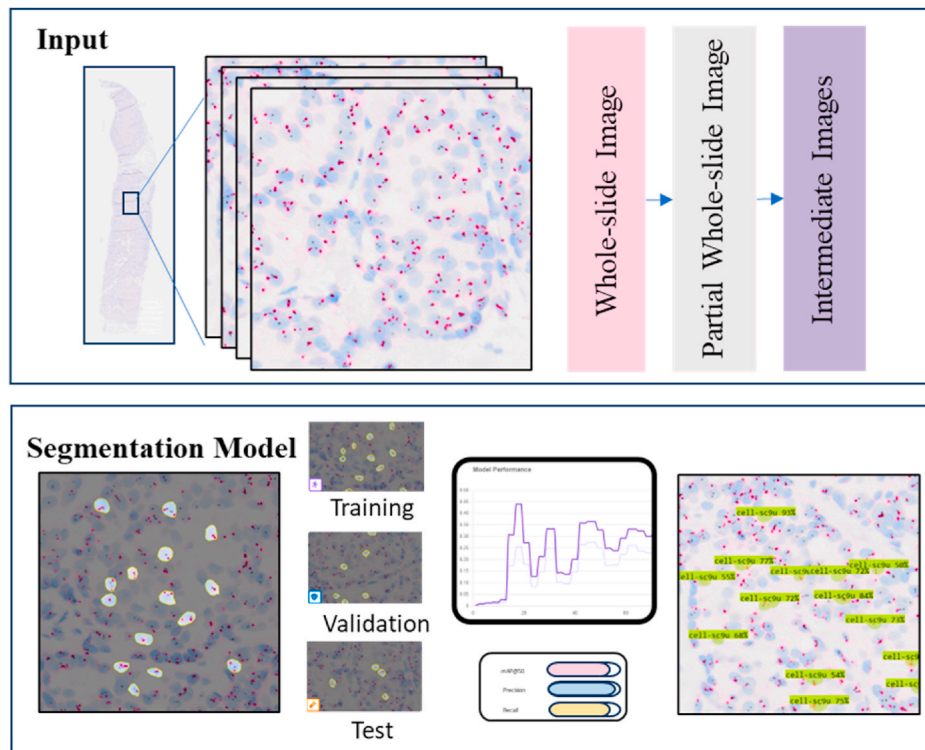


Fig. 4. Cancer cell detection user interface overview.

4. Result and discussion

4.1. Comparison models

This section compares two leading models—RF-DETR and YOLOv11—to highlight differences in architecture, accuracy, and efficiency. RF-DETR, a transformer-based model, emphasizes precision and data efficiency, while YOLOv11 focuses on real-time performance and lightweight deployment. Their key distinctions are summarized in Table VI.

The YOLOv11-seg model was employed to identify cellular regions, while the YOLOv11 model was utilized to detect red and black signal dots within the cells, which are critical for analyzing abnormal cellular characteristics. This combination successfully segmented and localized the marked regions, enabling clear distinction of signal-bearing cells from the background and surrounding tissue structures.

4.2. Segmentation performance

The three confusion matrices represent the segmentation performance of YOLOv11-fast, medium, and large models in detecting cell regions from Intermediate images. In Fig. 5, YOLOv11-fast achieved moderate sensitivity (18.0 % true positive) but suffered from a high false-negative rate (61.7 %), indicating under-segmentation of cells. The medium model improved true-positive detection to 26.2 % with a reduced false-negative proportion (42.3 %), showing better feature extraction and generalization. However, its false-positive rate (31.5 %) suggests occasional over-segmentation. The Large model maintained a similar true-positive rate (22.7 %) but increased false positives (37.4 %), implying higher sensitivity to noise and potential misclassification of background areas as cells. The predominance of false negatives highlights the intrinsic challenge of accurately identifying weakly stained or partially observable nuclei across morphologically heterogeneous tissue regions. The gradual increase in false positives from fast to large indicates that larger models may overfit on limited data.

Since each intermediate image contains a large number of cells, with some regions appearing blurred due to staining inconsistencies and illumination variations, the segmentation model was designed to identify and select only clear and well-defined cells for the next stage. The core function of segmentation model is to ensure that only high-quality cell regions are cropped and used as inputs for the HER2 and CEP17 object detection models. To achieve this, the YOLOv11-seg model employed a higher confidence threshold, ensuring that only cells with clear morphology and strong detection confidence were retained. From these results, selects the top 100 cells with the highest confidence scores for subsequent signal analysis. This approach enhances the reliability of medical interpretation by focusing on data quality rather than quantity. Among the YOLOv11-fast, medium, and large variants, the YOLOv11-medium model provides the balance between detection coverage and reliability. Thus, segmentation functions as a pre-filtering step, ensuring that only clear, diagnostically useful cells are passed to the downstream detection stage for red and black signal identification.

The clinical aspect of the model was examined as certified pathologists independently reviewed the detection results. They confirmed that the identified nuclei met the required criteria for HER2 DISH interpretation, including clarity, completeness, and proper chromogen staining. This expert validation supports the model's integration into semi-automated diagnostic pipelines, where reliable and consistent nucleus detection is essential for downstream HER2/CEP17 signal quantification. Moreover, the model exhibits rapid inference performance, making it suitable for practical deployment in high throughput pathology workflows without requiring manual intervention.

4.3. Object detection performance

The confusion matrices of the object detection performance of YOLOv11 (fast and medium) and RF-DETR (small and medium) models trained on the cropped cell images from the segmentation stage. The goal of this step was to detect black and red dots (representing HER2 and CEP17 signals) within each cell.

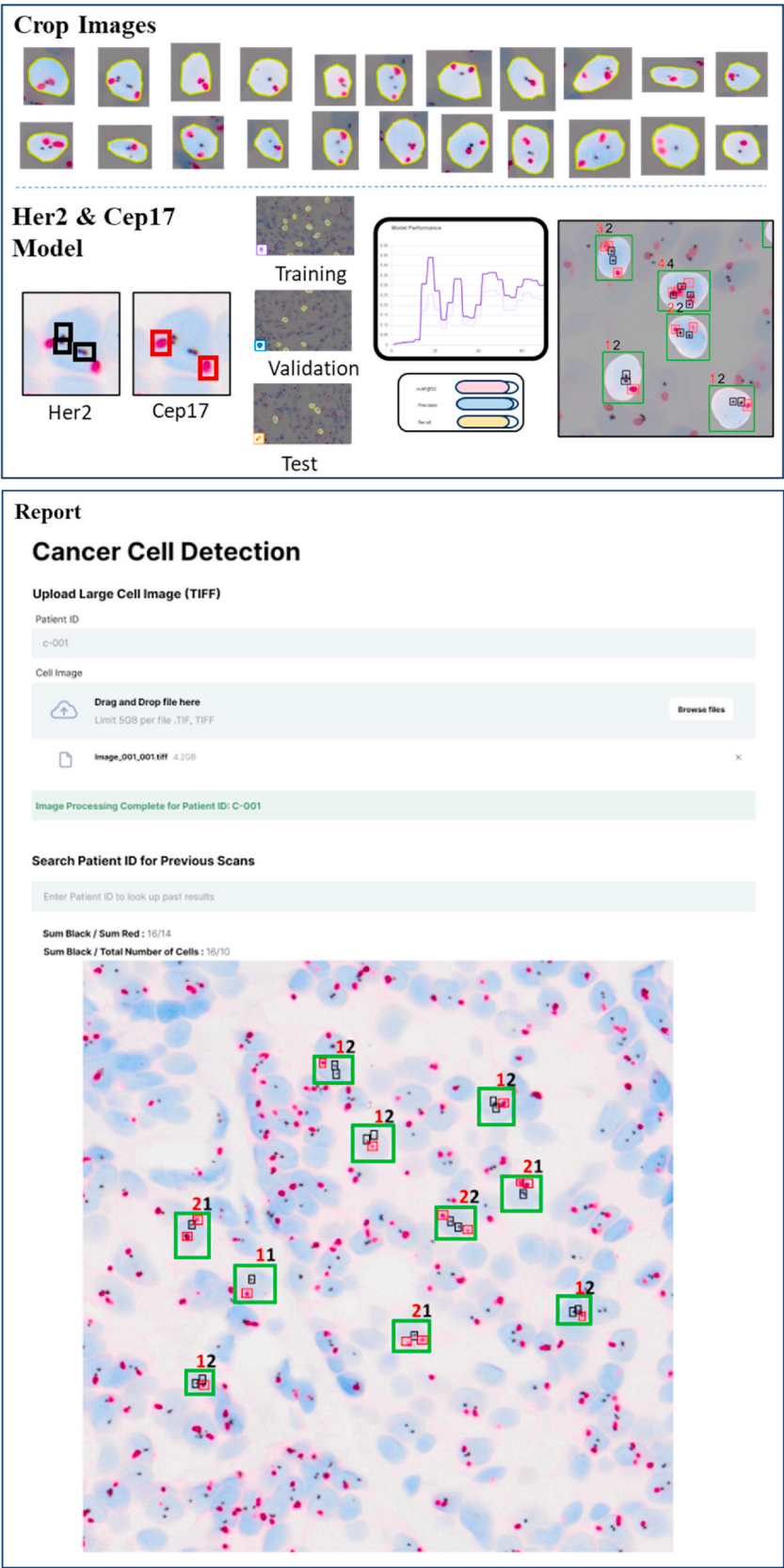


Fig. 4. (continued).

The comparison in the same model between the two confusion matrices YOLOv11-fast and YOLOv11-medium, shows that the YOLOv11-medium model performs substantially better than YOLOv11-

fast, particularly in detecting black (CEP17) signals. In the Fast model, there were 23 true positives for black signals (35.9 %) and 13 missed detections (20.3 %), while in the medium model, true positives

Table 6
Image augmentation for object detection models.

Feature	RF-DETR	YOLOv11
Architecture Type	Transformer-based (DETR family)	CNN-based (YOLO family, successor to YOLOv8)
Accuracy	Highest accuracy (especially on COCO benchmark)	Slightly lower accuracy on dense or small-object datasets
Speed	Real-time possible but slower than YOLO	Fastest inference, real-time capable
Training Efficiency	Slower training due to transformer computation	Efficient and easy to train
Data Efficiency	Needs less data, converges earlier	Needs more data to generalize well
Compatibility	Open Source (Apache 2.0)	Native Ultralytics ecosystem
Best Use Case	High-precision or dense-object detection	Real-time or embedded systems
Strengths	High precision Fewer false positives Works well with small datasets	Fastest inference Lightweight and flexible
Limitations	Slower training Heavier model size	Slightly less accurate on fine details

increased to 32 (49.2 %) and missed detections decreased sharply to only 4 (6.2 %). This improvement raised the recall for black signals from 63.9 % to 88.9 % and the F1-score from 76.5 % to 92.4 %, indicating that the medium model is much more sensitive to darker and low-contrast features. The detection of red (HER2) signals remained consistent between models, with 24 true positives (36.9 %) and recall of 92.3 % for both, showing that both configurations handle HER2 signals equally well. False positives and misclassifications were minimal and almost identical at 1.5 % in both cases. The medium model offers a superior balance between precision and recall, particularly enhancing detection reliability for the more challenging black signal class without sacrificing performance on red signals.

The confusion matrices of RF-DETR Small and RF-DETR medium models show highly consistent and accurate performance for detecting HER2 (red) and CEP17 (black) signals, with the medium version offering a slight but meaningful improvement. In the RF-DETR Small model, there were 33 true positives for black signals (52.4 %) and 25 true positives for red signals (39.7 %), with 3 missed black detections (4.8 %) and 1 missed red (1.6 %). The RF-DETR medium model increased the number of correct black detections to 35 (55.6 %), while reducing missed black detections from three to just one case (1.6 %), indicating better sensitivity. Both models maintained minimal misclassifications (only one black-to-red error) and showed no false positives, confirming excellent noise resistance and stable prediction behavior.

Quantitatively, both models achieved precision above 90 % for both black and red classes, indicating reliable and confident predictions with minimal background interference.

In conclusion, the RF-DETR medium model demonstrates slightly superior performance, especially in identifying black (CEP17) dots, while sustaining equally strong HER2 (red) signal detection. With precision exceeding 90 % and improved recall, the medium version offers a more balanced and dependable outcome for clinical HER2/CEP17 object detection tasks.

When compare between different model in most performance of each model. Both models deliver high precision and clinical reliability, but RF-DETR medium clearly outperforms YOLOv11-medium in terms of total accuracy, recall, and robustness, especially for CEP17 (black) signals that are often harder to detect. YOLOv11-medium remains advantageous for faster inference and computational efficiency, while RF-DETR medium provides the highest overall accuracy and consistency — making it the preferred model for precise clinical HER2/CEP17 analysis.

The YOLOv11-medium model achieved high, exceeding 95 %, demonstrating reliable performance in high-contrast signal detection. In comparison, the RF-DETR medium with precision above 90 % and zero

false positives. This indicates that RF-DETR medium not only identifies more signals correctly but also maintains exceptional detection stability across variations in dot intensity and size. While YOLOv11-medium provides faster inference and efficient HER2 detection, RF-DETR medium clearly outperforms it in overall detection accuracy, recall, and robustness—particularly for the more challenging black (CEP17) signals. Therefore, for clinical HER2/CEP17 analysis that demands both high precision and completeness, RF-DETR medium is the more reliable and diagnostically accurate choice.

4.4. Training performance assessment based on loss and mAP metrics

In Fig. 6, The first loss curve represents the segmentation model (YOLOv11-medium), while the second corresponds to the object detection model. These curves were obtained from the complete training dataset since the overall loss differences among data subsets were minimal. Therefore, a representative comparison between the two architectures was performed based on their full-data performance.

For the segmentation model, the loss values (Box, Class, and Object) exhibit small oscillations across epochs, indicating moderate convergence stability. The $mAP@0.5$ metric fluctuates between 0.3 and 0.5, suggesting that while the model effectively learns structural and boundary information, its segmentation precision varies slightly with complex cellular features.

In contrast, the object detection model shows rapid and stable convergence, with all loss components sharply decreasing within the first 50 epochs and maintaining low variance thereafter. The $mAP@0.5$ metric consistently rises to above 0.9, confirming strong localization and classification accuracy for both HER2 (red) and CEP17 (black) signals.

Overall, because the loss trends are not drastically different in magnitude, both models demonstrate adequate training stability. However, the object detection model achieves faster convergence and higher final performance, whereas the segmentation model retains slightly higher variability due to the complexity of pixel-wise feature extraction.

4.5. Color analysis by using the YOLOv11 object detection model

To enable accurate quantification of HER2 (black) and CEP17 (red) chromogenic signals, the proposed pipeline incorporates two specialized YOLOv11 object detection models. These models are applied to cropped nucleus images—previously segmented by the YOLOv11-seg model to independently detect signal points of each color. This automated detection replaces traditional manual counting, thereby improving diagnostic efficiency and minimizing observer dependent variability.

Experimental results show that both models accurately identify and count red and black signals within nuclei, with outcomes consistent with those obtained via standard pathological analysis. The quantitative outputs—namely, the HER2 and CEP17 signal counts—are directly used to compute the HER2/CEP17 ratio, which is essential for determining HER2 amplification status in accordance with clinical guidelines.

In Fig. 7, illustrates the output of the system at each stage: (a) nucleus segmentation, (b) HER2 signal annotation, (c) CEP17 signal annotation, and (d) the final merged detection result. These visualizations confirm that the proposed deep learning models effectively localize chromogenic signals in high resolution tissue images, supporting accurate and reproducible diagnostic interpretation.

4.6. Comparison of errors between the YOLOv11 model and manual counting

In Table VII, presents a comparative evaluation of HER2 (black) and CEP17 (red) chromogenic signal counts obtained using the proposed YOLOv11-based model versus manual counting by expert observers. The average error in red signal detection across four whole-slide images was 6.08 %, with individual deviations ranging from 5.05 % to 7.07 %. This

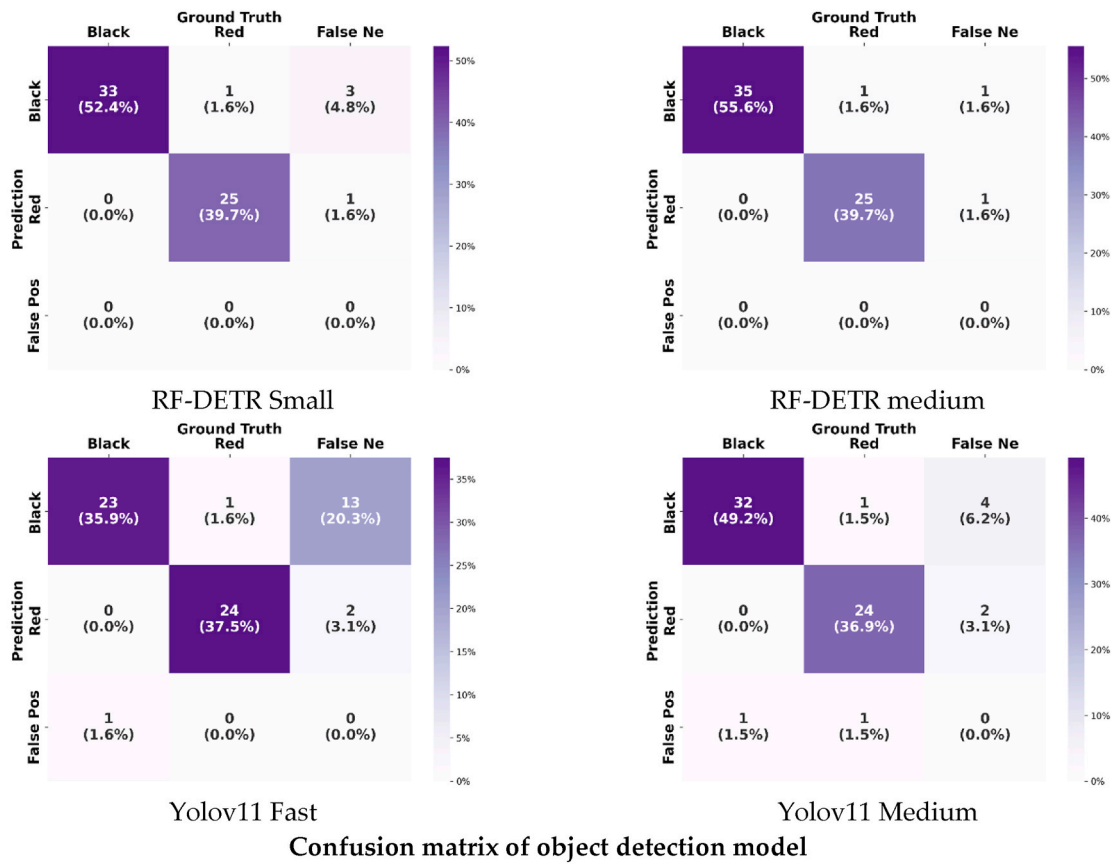
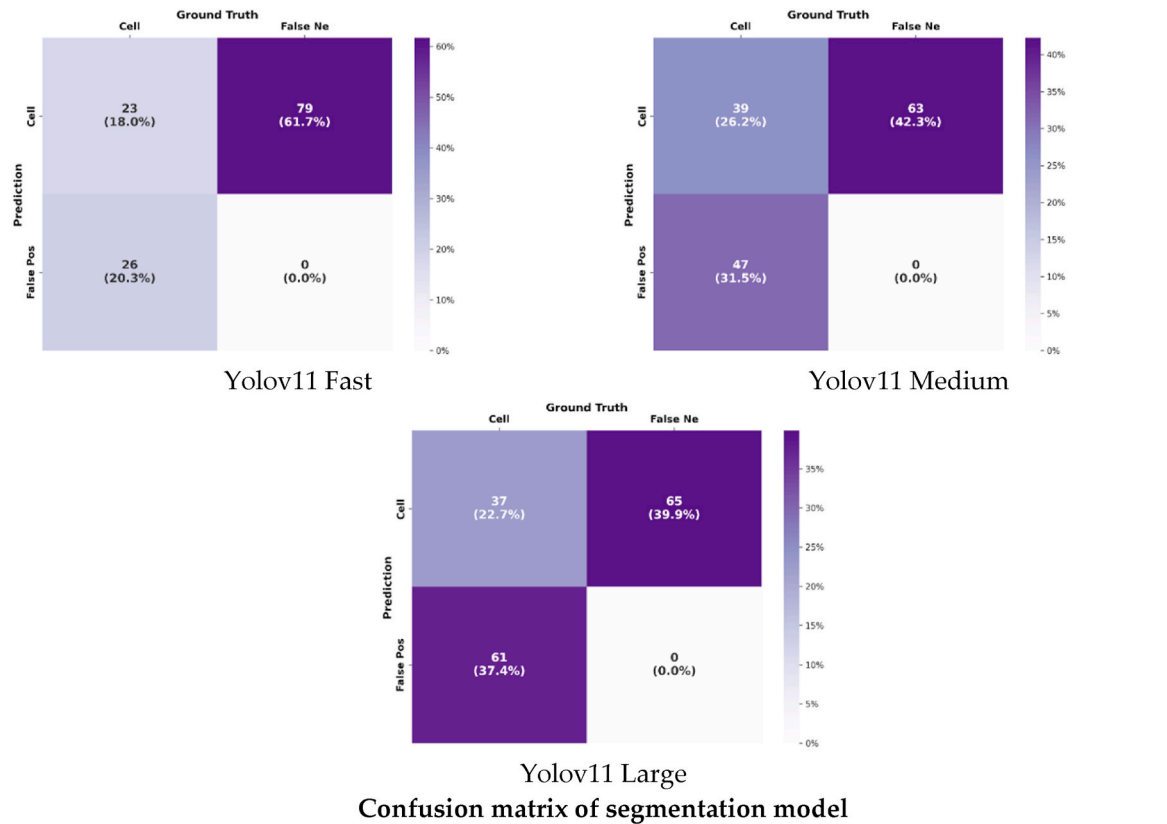
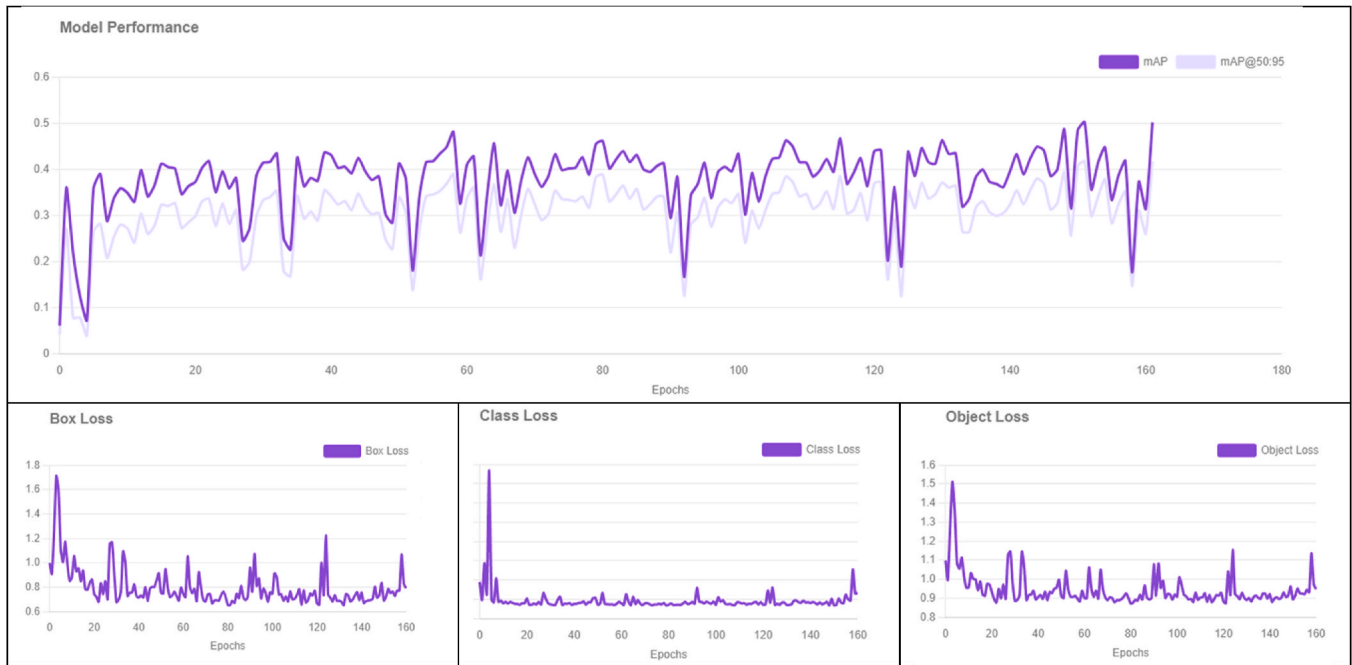
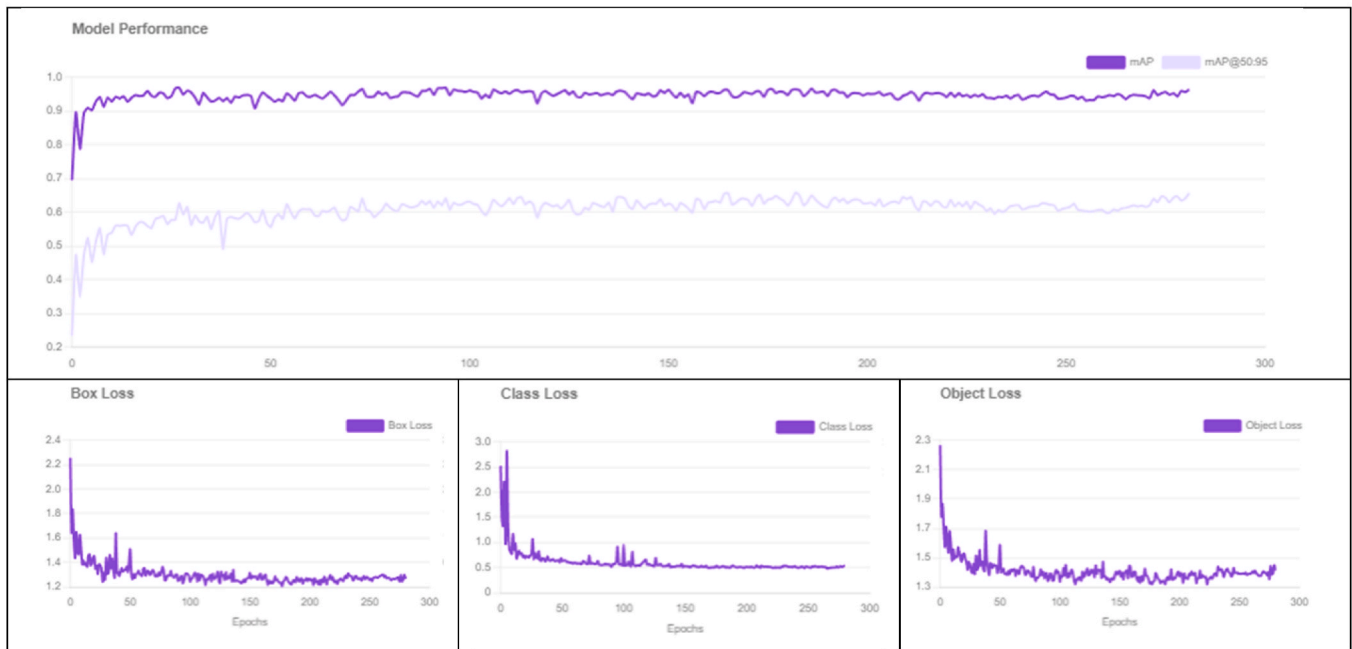


Fig. 5. Confusion Matrix for segmentation and object detection.



Loss curve of segmentation model Yolov11_Medium



Loss curve of Object detection model Yolov11_Medium

Fig. 6. Training Loss and Performance Curves of the models.

low and consistent error suggests that red signals—owing to their stronger chromogenic contrast and uniform shape—are more readily identified by the model.

In contrast, the average error in black signal detection was notably higher at 12.78 %, with the maximum deviation reaching 19.19 % in one of the images.

These elevated error rates are likely attributed several imaging and histological factors. First, black chromogenic signals often exhibit lower contrast against the hematoxylin-stained background, particularly in poorly stained or overexposed regions. Second, black signals frequently present with irregular morphology or fragmentation, which complicates their recognition by the model. Third, overlapping cells or dense signal

clusters may obscure discrete HER2 points, leading to either undercounting or merged detections.

Interestingly, in Image 4, the error for black signal detection dropped to 8.59 %, potentially reflecting improved lighting conditions or staining consistency. A qualitative review of failure cases revealed that the YOLOv11 model tends to overcount in areas where chromogenic precipitation creates high-density black noise or background artifacts. Conversely, undercounting occurs when signals are faint or partially occluded by adjacent structures.

Overall, the model shows higher reliability in detecting red chromogenic signals, highlighting a need for further optimization in black signal recognition. To address these limitations, several refinements are

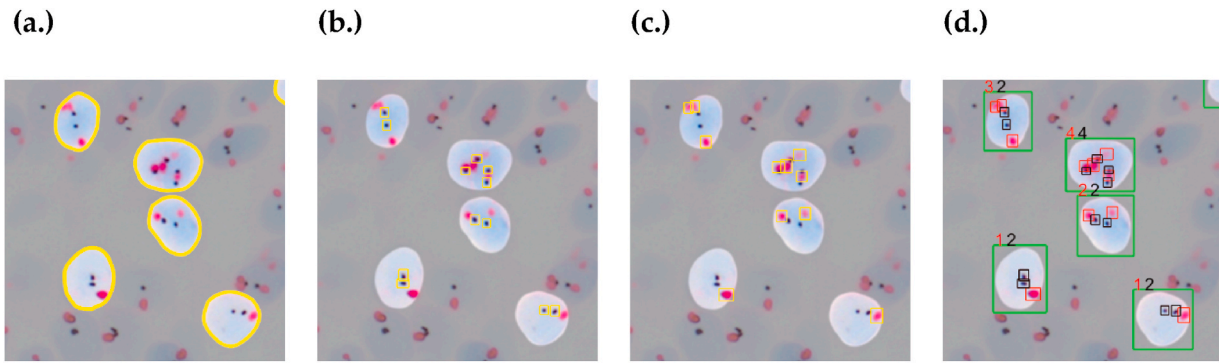


Fig. 7. Visual outputs of the proposed model: (a) Segmentation results, (b) HER2 annotation, (c) CEP17 annotation, and (d) final detection output.

Table 7

Comparison of error rates between YOLOv11 model and manual.

WSIs	the error of red points (%)	the error of black points (%)
1	6.63 %	12.24 %
2	5.56 %	19.19 %
3	5.05 %	11.11 %
4	7.07 %	8.59 %
Total	6.08 %	12.78 %

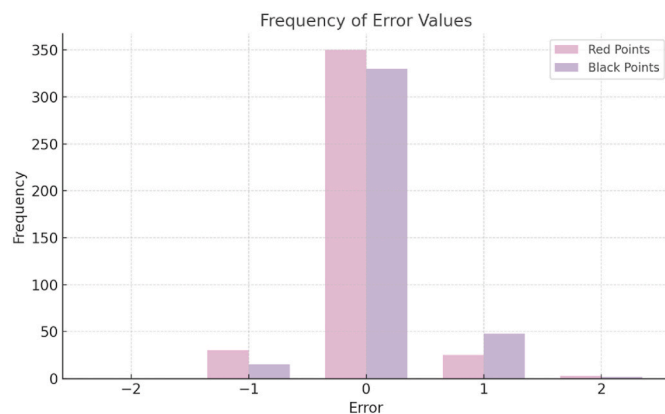


Fig. 8. Histogram of Counting Deviations: Red points (Cep17) Black point (HER2). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

proposed for future work. One approach is to apply color normalization or adaptive histogram equalization to enhance signal-background contrast prior to detection. Another promising strategy involves integrating attention-based mechanisms or multi-scale feature extraction within the object detection architecture, allowing better discrimination of subtle or overlapping signals. Additionally, employing a hybrid detection-classification model could filter false positives by validating detected points against learned shape and intensity priors. Incorporating these enhancements could improve black signal recognition robustness and narrow the performance gap between HER2 and CEP17 signal detection.

4.7. Statistical analysis of counting deviations

To quantitatively assess the performance of the YOLOv11-seg model in detecting and counting red (CEP17) and black (HER2) signals, a statistical comparison was conducted against manual annotations performed by expert pathologists. Two types of visual analyses—histograms and box plots—were employed to represent the distribution of counting deviations across four TIFF format whole-slide images. These

visualizations illustrate the frequency, central tendency, and dispersion of the prediction errors, providing insight into the model's accuracy and consistency under varying image conditions.

In Fig. 8, display histograms representing the distribution of counting deviations for red (CEP17) and black (HER2) points, respectively. The x-axis indicates the deviation values ranging from -2 to $+2$, while the y-axis shows the frequency of each deviation level across all evaluated cells.

For red points, the deviation values are highly concentrated around zero, indicating strong agreement between the model predictions and manual annotations. Over 90 % of deviations fall within the ± 1 range, reflecting high detection accuracy. The mean deviation was 0.002, with a standard deviation of 0.405, suggesting minimal variation and a consistently reliable output. This tight distribution implies that the system maintained stable performance under the observed staining and imaging conditions.

In contrast, black points exhibited a broader error distribution, with a noticeable skew toward positive deviations—particularly at $+1$ and $+2$. This indicates a moderate tendency for the model to slightly over-count HER2 signals, especially in regions with complex visual features or low staining contrast. Nevertheless, the model maintained an overall mean deviation of 0.094 and a standard deviation of 0.413. Importantly, 83.5 % of error values still clustered around zero, confirming that the system performed reliably across most scenarios. The presence of relatively few outliers further suggests that the detection algorithm operated under controlled and reproducible conditions, although further refinement may be necessary for improving precision in black signal detection.

To evaluate the practical utility of the proposed HER2DISH diagnostic system, a focus group comprising clinical pathologists and laboratory technicians was conducted following prototype deployment. Participants highlighted that the system produced clear and interpretable visual outputs, particularly beneficial in diagnostically ambiguous cases. Users reported improved clarity in HER2 signal interpretation and expressed confidence in the system's ability to support consistent decision making. Furthermore, the semiautomated workflow was praised for reducing diagnostic workload and streamlining slide review processes. Overall, the feedback from clinical personnel indicated that the system holds significant promise for integration into routine pathology practice and is viewed as a valuable decision support tool.

Statistical analysis revealed that red signal detection errors were predominantly centered around zero with minimal variance, whereas black signal detection exhibited a slightly wider error spread, indicating areas for potential model refinement. The model's occasional over-estimation of black points suggests that additional preprocessing or model training on varied staining conditions could enhance accuracy.

The integration of the entire diagnostic pipeline into a single application proved to be clinically meaningful. A focus group involving clinical pathologists and laboratory technicians confirmed that the system was intuitive to use, reduced diagnostic workload, and improved the consistency of HER2 evaluation. Notably, the automatic generation

Table 8
Comparison of manual Counting and Model based Top 100 cells selection.

Aspect	Manual Counting (Human Pathologist)	Model-Based Top-100 Cell Selection
Process	Pathologist manually selects and counts nuclei from each case (≈20–60 cells depending on clarity and borderline ratio).	Model automatically detects nuclei and ranks by segmentation confidence, selecting the top 100 well-segmented cells.
Typical Time (per case)	Few black dots (low HER2): 5–10 min Many black dots (high HER2): ~15 min Borderline cases (40–60 cells): >15 min	Automated selection: <3 min for cell detection and ranking. Manual verification (optional): 2–3 min
Cell Count Flexibility	Usually 20 cells for clear cases; increased to 40–60 cells if HER2/CEP17 ratio is borderline.	Fixed top-100 cells, ensuring statistical consistency and reproducibility across cases.
Consistency/Reproducibility	May vary between pathologists (subjective judgment).	High consistency due to standardized model-based selection and confidence-based ranking.
Accuracy/Reliability	High when done by expert pathologists, but subject to fatigue and inter-observer variability.	Comparable accuracy with reduced variability; depends on segmentation and detection model performance.
When Advantageous	Small number of cells (few HER2 dots). Quick screening cases.	Large number of cells or dots. Borderline or heterogeneous cases. When speed and objectivity are required.
Limitation	Time-consuming and potentially inconsistent across evaluators.	May occasionally include less optimal cells if confidence scores misrank due to image artifacts.

of HER2/CEP17 ratios and visual overlays helped mitigate human error and enabled faster case review. Participants noted that the system occasionally detected signals that were difficult to identify through manual inspection, supporting its role as a decision support tool.

These findings underscore the clinical relevance of incorporating AI-driven tools into routine diagnostic workflows. While traditional manual interpretation remains the gold standard, the proposed semiautomated pipeline offers a scalable solution for enhancing throughput and reliability in HER2 status determination. Future development will focus on improving the black signal detection model, expanding training datasets, and validating performance in larger multi institutional cohorts to further ensure generalizability. From a deployment perspective, the integrated pipeline was designed to run on standard workstation hardware (MSI Thin GF63 12ve-046 TH laptop equipped with a 12th Gen Intel (R) Core (TM) i7-12650H processor at 2.30 GHz), processing a single 1500 × 1500-pixel image in under 10 s, with end-to-end analysis of a batch of 100 nuclei completed in approximately 2 min. The standalone application does not require cloud computing or specialized GPU clusters, making it feasible for on-premise deployment in typical hospital IT environments. Future versions could support batch slide uploads and GPU-accelerated inference to further enhance scalability.

The future efforts will focus on improving black signal detection by expanding the training dataset and applying advanced preprocessing methods. Incorporating diverse samples and enhancing contrast and noise handling may improve accuracy. Clinical trials involving real-world diagnostic workflows will also be conducted to validate system usability and generalizability.

4.8. Comparison between manual counting and model-based cell selection

Assessing the practicality and clinical applicability of the proposed model-based Top-100 cell selection strategy, a comparative analysis was conducted against the traditional manual counting approach performed

by expert pathologists. This comparison highlights differences in workflow efficiency, reproducibility, and reliability under routine diagnostic conditions. In Table VIII, summarizes the key aspects of both methods, including their processes, time requirements, flexibility in cell selection, and overall consistency in HER2/CEP17 evaluation.

4.9. Comparison performance with other research

In this study, a dataset of 14,400 intermediate HER2-DISH images (1500 × 1500 px) was used to train and evaluate the proposed models. Each image contained clearly distinguishable HER2 (black) and CEP17 (red) chromogenic signals, representing realistic variations in clinical staining conditions. Among all tested architectures, the RF-DETR model achieved the highest overall detection accuracy of 95.24 %, outperforming other configurations of YOLOv11-seg and confirming its superior capability in detecting both red and black signal points with high precision and recall.

When compared with previous deep learning models, the performance of RF-DETR is comparable to or exceeds 90 % accuracy, placing it within the same high-performance range as HER2-ETNET (97.44 %) [14], SL-FCN (94.64 %) [28], and Vision Transformer (ViT) (98 %) [26]. However, the RF-DETR model in this research performs object-level signal detection, directly quantifying individual HER2 and CEP17 dots. This provides a more clinically interpretable output, enabling accurate HER2/CEP17 ratio computation consistent with diagnostic guidelines.

5. Conclusion

This study presents a clinically oriented, deep learning-based system for the automated analysis of HER2 gene amplification in breast cancer using Dual In Situ Hybridization (DISH) images. By integrating YOLOv11 and YOLOv11-seg models into a cohesive pipeline comprising cell detection, HER2/CEP17 signal quantification, and visualized reporting—this method achieves a balance between diagnostic accuracy and usability. Experimental evaluation shows a cell with HER2 and CEP17 signal The best-performing model achieved an accuracy of 95.24 % for the identification of HER2 and CEP17 signals, with quantitative analysis showing mean deviations of 12.78 % for HER2 and 6.08 % for CEP17, respectively, When compared to expert manual counts. Statistical analyses confirm the system’s robustness, particularly in red signal detection. The user friendly application supports routine clinical use, allowing pathologists to upload TIFF images, initiate analysis, and interpret results through intuitive visual outputs. Feedback from clinical users confirms the system’s diagnostic utility and its ability to reduce workload while maintaining precision. This work demonstrates the potential of deploying deep learning to support HER2 assessment in real-world pathology workflows.

For future work, further optimization of HER2 signal detection is needed, especially for black signals where variability remains higher due to visual ambiguity and background noise. Enhancing the dataset diversity and incorporating advanced preprocessing techniques may improve robustness. In addition, expanding the system to support whole-slide image analysis at scale and validating its performance in multi-center clinical settings will help ensure broader applicability and clinical readiness.

6. Ethics statement

This study was carried out with the formal approval and oversight of the Institutional Review Board (IRB), ensuring that all procedures involving human participants were conducted ethically, with careful attention to participant rights, safety, and confidentiality throughout the research process. The IRB approval was granted by the Institute of Pathology, Ministry of Public Health, Bangkok, Thailand (certification date: October 5, 2021). The titled ‘Multi Artificial Intelligence (AI

Development in Digital Pathology' (Project No. IOP-KM_R65-001).

CRedit authorship contribution statement

Terisara Micaraseth: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Shanop Shuangshoti:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Investigation, Data curation, Conceptualization. **Sakdina Prommaouan:** Supervision, Resources, Investigation, Formal analysis, Data curation. **Somruetai Shuangshoti:** Supervision, Resources, Investigation, Formal analysis, Data curation. **Rizwan Ullah:** Writing – review & editing, Supervision, Conceptualization. **Gridsada Phanomchoeng:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT (GPT-5, OpenAI) in order to assist in English language editing and refinement of academic writing. After using this tool/service, the author (s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to express their sincere gratitude to Dr. Worapat Teachasupinun, Pathologist at King Chulalongkorn Memorial Hospital, for his professional guidance in the field of pathology. His valuable insights and thoughtful feedback were essential to the development and performance evaluation of the image analysis. And gratefully acknowledge from Mr.Hattakit Jarusan, Mr.Kolawat Dendeevanich, and Mr.Thanabordee Luangasnathip for providing valuable information and insights that supported the completion of this research.

This research project is supported by the Second Century Fund (C2F), Chulalongkorn University, Bangkok, Thailand; National Research Council of Thailand (NRCT) under Grant N42A670553; 111th Anniversary Engineering Research Catalyst Fund Towards U Top 100; and Program Management Unit for Human Resources and Institutional Development, Research and Innovation under Grant B05F640233.

References

- [1] Sung H, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2021;71(3): 209–49.
- [2] Society, A.C.. Cancer facts & figures 2024. In: American cancer society. Atlanta, GA: American Cancer Society; 2024.
- [3] Society, A.C.. What causes breast cancer? Atlanta, GA: American Cancer Society; 2021.
- [4] Edelweiss M, et al. HER2 assessment by bright-field dual in situ hybridization in cell blocks of recurrent and metastatic breast carcinoma. *Cancer cytopathology* 2019;127(11):684–90.
- [5] Lander EM, et al. Using the HER2/CEP17 FISH ratio to predict pathologic complete response following neoadjuvant anti-HER2 doublet therapy in HER2+ breast cancer. *Oncologist* 2023;28(2):123–30.
- [6] Anekpuritanang T, Chaichana T, Ruangvejvorachai P, Shuangshoti S. Reproducibility in the assessment of HER2 DISH in breast cancer. *The Bangkok Medical Journal* 2015;10. pp. 1–1.
- [7] Society, A.C.. Breast cancer basics. Atlanta, GA: American Cancer Society; 2022.
- [8] Elmore JG, et al. Variability in interpretive performance at screening mammography and radiologists' characteristics associated with accuracy. *Radiology* 2009;253(3):641–51.
- [9] Pathologists, C.o.A.. Invasive breast cancer biomarker reporting and testing. Northfield, IL: College of American Pathologists (CAP); 2021.
- [10] National Cancer Institute, M.o.P.H., Thailand, National cancer screening guidelines 2024. National Cancer Institute, Ministry of Public Health: Nonthaburi, Thailand.
- [11] Nitta H, et al. Development of automated brightfield double in situ hybridization (BDISH) application for HER2 gene and chromosome 17 centromere (CEN 17) for breast carcinomas and an assay performance comparison to manual dual color HER2 fluorescence in situ hybridization (FISH). *Diagn Pathol* 2008;3:1–12.
- [12] Bannier PA, et al. Development of a deep-learning model tailored for HER2 detection in breast cancer to aid pathologists in interpreting HER2-low cases. *Histopathology* 2024;85(3):478–88.
- [13] Esteve A, et al. A guide to deep learning in healthcare. *Nat Med* 2019;25(1):24–9.
- [14] GP, P., K. Senapati, and A.K. Pandey, Improving the performance of multi-stage Her2 breast cancer detection in hematoxylin-eosin images based on ensemble deep learning. Available at: SSRN 4673047.
- [15] Silva HECd, et al. The use of artificial intelligence tools in cancer detection compared to the traditional diagnostic imaging methods: an overview of the systematic reviews. *PLoS One* 2023;18(10):e0292063.
- [16] Redmon J, Divvala S, Girshick R, Farhadi A. You only look once: unified, real-time object detection. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2016.
- [17] Litjens G, et al. A survey on deep learning in medical image analysis. *Med Image Anal* 2017;42:60–88.
- [18] Zhu Z, Sun Y, Honarvar Shakibaei Asli B. Early breast cancer detection using artificial intelligence techniques based on advanced image processing tools. *Electronics* 2024;13(17):3575.
- [19] Jiang X, Hu Z, Wang S, Zhang Y. Deep learning for medical image-based cancer diagnosis. *Cancers* 2023;15(14):3608.
- [20] Bi WL, et al. Artificial intelligence in cancer imaging: clinical challenges and applications. *CA Cancer J Clin* 2019;69(2):127–57.
- [21] R T, V.K. V M, Guluwadi S. Enhancing brain tumor detection in MRI images through explainable AI using Grad-CAM with resnet 50. *BMC Med Imag* 2024;24(1):107.
- [22] Bochkovskiy A, Wang C-Y, Liao H-YM. Yolov4: optimal speed and accuracy of object detection. arXiv preprint arXiv:2004.10934 2020.
- [23] Lu MY, et al. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nat Biomed Eng* 2021;5(6):555–70.
- [24] Yang J, et al. Prediction of HER2-positive breast cancer recurrence and metastasis risk from histopathological images and clinical information via multimodal deep learning. *Comput Struct Biotechnol J* 2022;20:333–42.
- [25] Wang S, et al. AWCDL: automatic weight calibration deep learning for detecting HER2 status in whole-slide breast cancer image. *Intelligent Oncology* 2025;1(2): 128–38.
- [26] Rehman ZU, et al. Deep-learning-based approach in cancer-region assessment from HER2-SISH breast histopathology whole slide images. *Cancers* 2024;16(22):3794.
- [27] Bappi JO, et al. A novel deep learning approach for accurate cancer type and subtype identification. *IEEE Access* 2024.
- [28] Wang C-W, et al. A soft label deep learning to assist breast cancer target therapy and thyroid cancer diagnosis. *Cancers* 2022;14(21):5312.
- [29] Wolff AC, Somerfield MR, Dowsett M, Hammond MEH, Hayes DF, McShane LM, Saphner TJ, Spears PA, Allison KH. Human epidermal growth factor receptor 2 testing in breast cancer: ASCO-college of American pathologists guideline update. *J Clin Oncol* 2023;41:3867–72.